A Two-Stage Open Compound Domain Adaptation Framework for Semantic Segmentation in Remote Sensing Imagery

Zhi Gao, *Member, IEEE*, Ziyao Li, Mengjie Xie, *Graduate Student Member, IEEE*, and Qiao Wang

***Abstract*— Unsupervised-domain adaptation (UDA) has emerged as a critical research direction in remote sensing (RS) interpretation, aiming to bridge the gap between the labeled source and unlabeled target domains. However, most methods are designed for either a single target domain or multidomain setting with clear boundaries, which necessitates retraining UDA models for each target domain, making it even harder to directly generalize to unseen domains. This article proposes a novel two-stage open compound domain adaptation (OCDA) framework for semantic segmentation in RS images, which models the target domain as a composite of multiple unknown but homogeneous domains and leverages image translation and meta-learning techniques. In the first stage, we meticulously design a cross-domain image translation (CDIT) model based on contrastive learning to rapidly align the appearance of target-domain images with the source-domain style. In the second stage, the translated target images are first processed by a pretrained model to generate pseudo-labels. Subsequently, a dynamic classwise memory (DCWM) model is designed to progressively update abstract categorical features, serving as external class guidance for semantic segmentation within a meta-learning framework. Specifically, meta-training is employed to iteratively learn and update domain-agnostic categorical memory of semantic classes, while meta-testing simulates memory retrieval and gradually refines the categorical memory using pseudo-labels to adapt to new domains. Additionally, intraclass cohesion and interclass divergence losses are incorporated to enhance the abstraction and retention of categorical features, aligning more closely with human cognitive patterns. Extensive experiments on RS benchmarks and unseen real images demonstrate the superior generalization of our method compared to the state-of-the-art approaches.**

***Index Terms*— Image translation, meta-learning, remote sensing (RS), semantic segmentation, unsupervised-domain adaptation (UDA).**

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I. Introduction

NOWADAYS, the deep learning-based semantic segmentation models have shown promising results in the intelligent interpretation of remote sensing (RS) images, taking advantage of large datasets and sophisticated network architectures. These advancements are crucial for enhancing human understanding and exploration of the Earth's surface, particularly in supporting natural disaster monitoring, geographic information mapping, and so on. [1], [2]. However, in practical applications, RS images encompass broader scenes than daily images, with their appearance distributions influenced by a wider range of factors, including different sensors, spectral bands, geographic locations, scales, and seasonal changes [3], as shown in Fig. 1(a). This naturally results in the same object with different spectra, and different objects with the same spectrum. Such diversity poses a substantial challenge for data-driven deep learning models, which are trained on a specific dataset (source) and often fail to generalize well to new, unseen domains (target). This phenomenon, known as domain shift, is particularly pronounced in RS imagery. Additionally, for each new domain, manually annotating labels for target data is impractical.

Unsupervised-domain adaptation (UDA) has been proposed to address performance degradation caused by domain shifts, providing a practical way to leverage labeled source data and unlabeled target data without additional manual labeling efforts. Research on UDA for RS typically focuses on aligning the source and target domains to a unified image style [4], [5] or feature space [6]. Despite achieving impressive results, the existing methods still face limitations when applied to complex real-world scenarios. First, most existing works are confined to fixed-domain adaptation settings, where one source domain corresponds to one target domain. This constraint significantly hinders the scalability of domain adaptation models, as they require network retraining or fine-tuning for new target domains, not to mention the formidable challenge of generalizing to "any" unseen domain. Second, in a minority of studies on multitarget-domain adaptation, it is usually assumed that well-defined boundaries exist among the target domains [7], [8], enabling their initial classification prior to domain adaptation. However, in practice, it is challenging to define such clear, known distinction boundaries between domains, and test images sometimes incorporate unseen domains. It is

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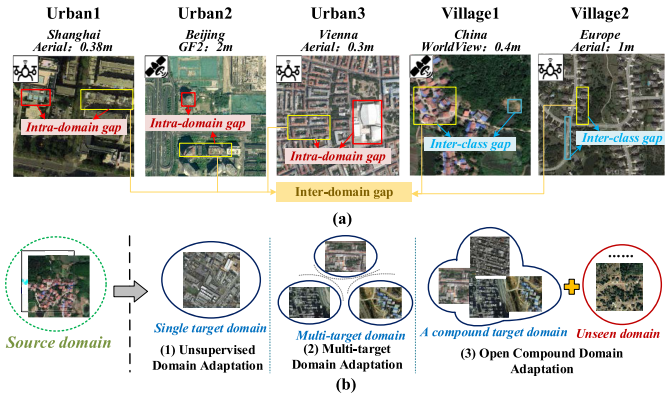


Fig. 1. (a) Intuitive cases of domain gaps in RS images semantic segmentation. (b) Illustration of different UDA settings. The open compound target domain comprises multiple traditionally homogeneous domains and allows novel domains to appear during inference.

impractical to separate the data into discrete domains for domain adaptation.

These challenges prompt the research presented in this article. We comprehensively evaluate domain adaptation methods for deep learning-based semantic segmentation models. Starting from real intelligent interpretation scenarios, we develop a two-stage open compound domain adaptation (OCDA) [9], [10] framework under a domain setting that more closely aligns with real-world conditions. Fig. 1(b) compares the different domain adaptation settings. Our task is to train a model using labeled source-domain data and adapt it to the unlabeled compound target domain, which consists of multiple traditionally homogeneous domains, while also generalizing to unseen domains.

We tackle this challenge from two key perspectives. In the first stage, we design a cross-domain image translation (CDIT) model to fully utilize labeled source data and pretrained models, reducing appearance discrepancies between target datasets and the source dataset through style transfer. Notably, the RS cross-domain datasets used to train the style transfer model typically lack one-to-one correspondences, making it harder to ensure semantic consistency [3]. Traditional style transfer methods, which rely on perceptual losses and Gram loss for style alignment, often fall short in maintaining semantic consistency and preserving image details and even suffer from overstylization [11], [12]. Therefore, we design a CDIT network for unpaired images, which leverages contrastive learning [13] to extract comparable information from a single image for unsupervised style transfer. Specifically, we achieve rapid alignment of appearance through a style contrastive loss, while preserving the original image details with a content contrastive loss. Here, the source-domain style serves as a unified style for subsequent domain adaptation learning.

Moreover, the rich variety of land cover samples in the compound target domain enables the model to learn the diversity of categorical features across different domains, which is crucial for mitigating overfitting to the source domain. Therefore, in the second stage, we integrate pseudo-labeling and meta-learning to achieve adaptation to the target and unseen domains. Specifically, the translated target images are initially processed by a pretrained model to generate pseudo-labels, followed by their filtering based on entropy confidence scores. Additionally, conceptual knowledge, also known as semantic memory [14], is abstracted from real-world experiences and generalized into reusable forms to support new cognitive activities [15]. Inspired by this, we design a dynamic classwise memory (DCWM) model in the feature space to extract and store common categorical features across domains. Concretely, we introduce intraclass cohesion and interclass divergence losses to enhance domain-invariant representations and to boost the abstraction and memorization of categorical features, respectively. We also leverage meta-training and meta-testing processes to update and read domain-invariant generalized memory features of different classes, rather than simple feature alignment. During inference on unseen open domains, this model utilizes an external fixed memory matrix as class guidance to reduce ambiguity in the representations of domain test data, thereby enhancing its adaptability to novel domains.

Our contributions are summarized as follows.

1) We propose a novel framework for unsupervised OCDA specifically designed for high-resolution RS images, which differs from one-to-one domain adaptation and multitarget-domain adaptation. Our method integrates an image translation stage with a meta-learning-based categorical memory learning stage. This design facilitates adaptation to the target domain and improves generalization to unseen open domains.
2) We design a contrastive learning-based image translation model CDIT to bridge the appearance gap between datasets and generate high-quality pseudo-labels. The elaborately designed patch-based multilayer content and style contrastive losses are employed to ensure consistency between the translated and original images, while rapidly aligning their style features.
3) We present a dynamic categorical memory guidance (CMG) algorithm within a meta-learning framework to exploit and store domain-agnostic categorical knowledge. By testing and optimizing the model with pseudo-labels in the compound target domain, we improve the representation power of the memory-guided features, enabling the model to generalize to open unseen domains.

II. Related Works

A. UDA and Generalization

Extensive research has been conducted toward better generalization of deep networks to mitigate the performance degradation caused by domain shifts. In this part, we review the literature on UDA and domain generalization (DG). Specifically, UDA methods aim to rectify such domain mismatches by utilizing labeled source images and unlabeled target images [16]. Representative methods include latent distribution alignment [17], adversarial discrimination [18], self-training [19], and cycle consistency [20]. However, the early domain adaptation settings typically focus on the "one source domain to one target domain" [9] and are not equipped to handle more complex scenarios involving multiple

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target domains. Consequently, multitarget-domain adaptation has been introduced to address more practical scenarios, extending UDA to multiple [21] or continuous [22] target domains. Nevertheless, these methods usually assume a clear domain distinction or require domain labels [9], which are rarely feasible in real-world scenarios. In view of these problems, OCDA has been proposed to harness the annotated source images and the unannotated target images captured in open environments to train the network. It is worth noting that OCDA differs from DG [15], [23], with DG posing a greater challenge due to the unavailability of target-domain images. In this work, we leverage the OCDA settings and extend them to unseen domains, which is more applicable to real-world RS scenarios.

B. UDA in RS

The current dedicated research on UDA for RS imagery remains inadequate, and the potential of cross-domain adaptation has not been fully exploited [24]. Adapting from a single source domain to various target domains poses considerable challenges, let alone transferring to unseen domains. Despite this, numerous UDA methods for RS have still been proposed. We categorize these methods into four main types: appearance-level alignment, feature-level alignment, self-training, and hybrid methods. Appearance-level alignment methods aim to align the color or spectral distribution between the source and target images by minimizing the appearance differences [25]. Target-domain images are converted to source-domain style [3], [5] or a standardized style [4], [26] before being fed into a pretrained model for prediction. The most common technique involves using image-to-image (I2I) translation networks [20] or color mapping methods [27] for image alignment. However, these methods may struggle to ensure the semantic consistency and tonal authenticity of the translated images. Feature-level alignment methods aim to achieve domain confusion by designing a domain discriminator [28] that distinguishes multilevel domain-invariant features across domains. However, these methods potentially lead to negative transfer to the target domain due to overfitting to the source-domain features. Self-learning-based methods leverage pseudo-labels generated by models trained on the source domain and utilize various training strategies, such as contrastive learning [29] and transformer [30] to train or fine-tune the target-domain model. However, the pseudo-labels employed for supervised learning are inherently imprecise, which can propagate and accumulate errors during the training process. Considering the respective advantages and limitations of each of the above approaches, a current trend in RS is to combine them [25], [31], [32]. Moreover, most existing RS UDA methods rely on the assumption of a "single-source to single-target" scenario, necessitating retraining or updating the domain adaptation model for new target domains, thereby limiting their generalization capability.

C. Style Transfer Networks

The term "style" originally refers to the painting styles of different artists. Nowadays, it is used to describe the different appearances of images [33]. Style transfer methods generate new images in the style of a particular artist by rendering an input content image using style information extracted from an image of a real artwork. Due to their powerful ability to abstract features, these methods often result in detail loss and are unsuitable for the photorealistic transfer. In the RS field, "style" refers to variations in appearance caused by factors, such as different geographic locations, seasons, and sensors. Style transfer between images must adhere to the principles of semantic consistency and detail preservation. However, in practice, many RS image alignment methods still utilize low-order statistical value matching techniques, such as histogram matching [34] and MKL [27] for color alignment between images. Unfortunately, they mainly focus on color matching in images and usually ignore the semantic logic. Fortunately, I2I translation [20] and photorealistic style transfer [35] methods have provided more promising approaches for style alignment in RS images. Specifically, the mainstream deep learning-based methods employ the architecture of CycleGAN [20] and its variants [36], utilizing adversarial loss to constrain image style and cycle-consistency loss to preserve content. However, these methods still suffer from overstylization, resulting in distortion of image details. Furthermore, cycle consistency assumes a bijective mapping between the two domains, which is often too restrictive. Real cross-domain RS images rarely have one-to-one co-responding mappings. Hence, employing unsupervised constraints on unpaired datasets while maintaining consistency in both semantics and content details is a significant challenge.

D. Meta-Learning and Memory Networks

Meta-learning, also known as learning-to-learn, uses previous knowledge and experiences to guide new task learning, equipping networks with the ability to learn [37]. Meta-learning has garnered significant interest in various fields, including transfer learning, domain adaptation, DG, continual learning, and multitask learning [38]. Memory-augmented neural networks [39], a highly capable meta-learning method that utilizes explicit storage buffers [40], have demonstrated strong performance in domain adaptation. Sukhbaatar et al. [41] used an external memory unit in semi-supervised semantic segmentation and DA. Kim et al. [15] proposed a meta-learning method to store long-term memory throughout the training process, which helps to robustly read and write memory for UDA. Su et al. [42] leveraged few-shot learning to mitigate cross-domain distribution discrepancies and enable domain adaptation for novel classes. Su et al. [43] introduced a meta-learning framework incorporating meta-batch normalization and a distance constraint to enhance the generalization ability of diagnostic classification models for medical image analysis. Furthermore, Chen et al. [29] enhance interclass distinguishability by storing class features and per-pixel features from the labeled source-domain data in a memory module. Compared to previous works, we not only record category features from labeled data but also store domain-agnostic information from unlabeled target data. This approach introduces auxiliary memory to store and reuse common semantic features, facilitating unseen domain adaptation.

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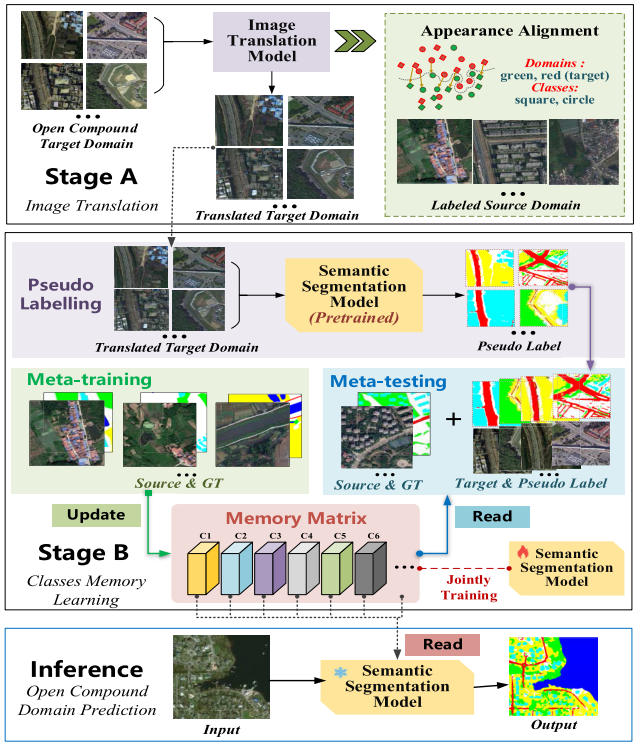


Fig. 2. Illustration of the proposed method. Stage A focuses on aligning images using an image translation network, while Stage B generates pseudo-labels through a pretrained model and then learns the dynamic categorical memory-guided model. The fully trained model is then capable of performing inference on images from unseen domains.

III. Methodology

In this section, we explain the design philosophy of our approach. A schematic overview of the entire approach is shown in Fig. 2. The contents of this section are as follows: we begin with an overview of the overall architecture, followed by a detailed explanation of the two stages of the framework.

A. Overview of the Proposed Method

1) Preliminaries: Unlike naive UDA, which aims to improve model performance on a specific target domain, and DG, which enhances performance on unseen domains using only source-domain data, our approach jointly trains on both labeled source domain and unlabeled target domain, effectively mitigating overfitting to the source domain while enhancing the model's generalization to unseen domains.

First, we define some notations used throughout this article. The labeled source domain is defined as $S = \{(x_s, y_s)|x_s \in \mathbb{R}^{H_S \times W_S \times 3}, y_s \in \mathbb{Z}^{H_S \times W_S \times N}\}$ with a resolution of r_S, where x_s is the source image, y_s is the corresponding semantic annotation, and N is the number of classes. For open compound target domain T, which comprises target images x_T^i from multiple homogeneous subtarget domains, each subtarget domain is denoted as $T^i = \{x_T^i | x_T^i \in \mathbb{R}^{H_T \times W_T \times 3}\}$, $i = 1, \ldots, D$, with a resolution of r_T^i, where D is the number of homogeneous subtarget domains. These subtarget domains are considered unknown and lack clear boundaries, collectively forming a large compound set. Thus, we denote target images as x_T. The aim of this article is to develop a domain adaptation semantic segmentation model $\boldsymbol{F}_{SS}: x \to \mathrm{y} \in \mathbb{Z}^{H \times W \times N}$, which can generate prediction labels y for images x from the target and unseen domains.

As illustrated in Fig. 2, our proposed method consists of two stages: the image translation based on contrastive learning and the CMG based on meta-learning. Stage A primarily carries out style transfer between unpaired datasets. Concretely, a CDIT model $\boldsymbol{I}_{\mathrm{CDIT}}: T \to S$ based on contrastive learning is developed to generate translated images T' of the open compound target domain in the style of the source domain S. Stage B involves two steps. First, the translated images T' are input into a pretrained semantic segmentation network, originally trained on source-domain data, to generate the corresponding pseudo-labels $\hat{y}$. Then, a DCWM model $\boldsymbol{M}_{\mathrm{DCWM}}: x \to \mathcal{M}$ undergoes progressive learning through alternating meta-training and meta-testing phases. In this stage, the memory model $\boldsymbol{M}_{\mathrm{DCWM}}$ is integrated with the segmentation network $\boldsymbol{F}_{SS}$ to store domain-agnostic categorical features of different semantic classes in a memory matrix $\mathcal{M} \in \mathbb{R}^{N \times C}$, where N is the number of classes and C is a channel dimension of the encoder feature. Within the meta-learning framework, we update and read the memory matrix using source images with ground truth and target images with pseudo-labels as guidance.

2) Scale Problem: Scale diversity remains a challenging issue for the intelligent interpretation of RS imagery. This study specifically focuses on cross-domain adaptation for multiclass semantic segmentation of very high-resolution ($r_S, r_T < 1$ m) visible images. Medium-resolution and low-resolution images fall outside the scope of our study. Additionally, to mitigate the impact of scale factors on model training, the initial image patch size ($H_S \times W_S$, $H_T \times W_T$) should be cropped based on the image resolution r_S, r_T according to the following formulation:

$$\frac{r_S}{r_T} = \frac{H_T}{H_S} = \frac{W_T}{W_S}. \tag{1}$$

B. Stage A: Image Translation Based on Contrastive Learning

1) CDIT Model: The essential goal of CDIT is to learn the mapping functions $\boldsymbol{I}_{\mathrm{CDIT}}: T \to S$, generating translated target images T' with the style of domain S, thereby improving the quality of pseudo-labels in the second stage. The CDIT network, as shown in Fig. 3(a), adopts a GAN architecture with a generator and a discriminator. Specifically, we utilize Resnet_9blocks [44] and DeeplabV3+ [45] as the backbone for the encoder of $\boldsymbol{I}_{\mathrm{CDIT}}$ and the segmentation model, respectively. For the discriminator $\boldsymbol{D}$, we employ the multiscale architecture from [46] to ensure that the output closely resembles images from the domain S. Contrastive learning captures object relationships by learning the similarities between inputs within the same distribution, i.e., associating data points, called "queries," with their "positive" examples, while distinguishing them from "negative" examples. We transform the problem of maximizing mutual information into an $(N+1)$th-way classification problem similar to [13], where the distances between queries and other examples are scaled by a temperature parameter $\tau = 0.07$. The cross-entropy loss is then

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calculated as follows:

$$\ell\left(\boldsymbol{q}, \boldsymbol{k}^{+}, \boldsymbol{k}^{-}\right) = -\log\left[\frac{\exp\left(\boldsymbol{q}\cdot\frac{\boldsymbol{k}^{+}}{\tau}\right)}{\exp\left(\boldsymbol{q}\cdot\frac{\boldsymbol{k}^{+}}{\tau}\right)+\sum_{n=1}^{N}\exp\left(\boldsymbol{q}\cdot\frac{\boldsymbol{k}_{n}^{-}}{\tau}\right)}\right] \tag{2}$$

where $\boldsymbol{q}, \boldsymbol{k}^{+} \in \mathbb{R}^{K}$ denote the K-dimensional vectors ("queries" and "positive" keys, respectively), and $\boldsymbol{k}^{-} \in \mathbb{R}^{N\times K}$ represents a set of N "negative" K-dimensional vectors.

In this article, we design two types of contrastive losses: style contrastive loss and content contrastive loss, with their patchwise architecture depicted in Fig. 3(b).

2) Style Alignment Constraints: We propose a style contrastive loss to encourage the model to learn toward the target style from the perspective of unsupervised training. Park et al. [13] demonstrate that applying contrastive learning in a multilayer, patchwise paradigm extracts positive and negative examples more effectively from a single image itself than from other images in the dataset. This approach opens up the possibility for one-shot image translation in the future.

We select L-layer feature maps from the encoder of $\boldsymbol{I}_{\text{CDIT}}$ and pass them through a style projection network $\boldsymbol{H}_S$, which is a two-layer MLP with 128 units in each layer. Here, the contrastive vector is a stack of features denoted as follows:

$$\{\boldsymbol{s}_l(x)\}_L = \left\{\boldsymbol{H}_S\left(\boldsymbol{I}_{\text{CDIT}}^{\text{enc}-l}(x)\right)\right\}_L \tag{3}$$

where x is the encoding image and $\boldsymbol{I}_{\text{CDIT}}^{\text{enc}-l}$ represents the output of the lth selected layer. We refer to the random feature as $\boldsymbol{s}_l^r(x)$, $r \in \{1, \ldots, R_l\}$, where R_l is the number of patches in each layer, set to 256. Then, we put the fake $x_T{}'$ and real x_T and x_S into (3) to get the "queries" $\boldsymbol{s}_l^r(x_T{}')$, "negative" examples $\boldsymbol{s}_l(x_T)$, and "positive" examples $\boldsymbol{s}_l^r(x_S)$, respectively. Thus, the style contrastive loss is defined as follows:

$$\mathcal{L}_{\text{S}-\text{contra}} = \sum_{l=1}^{L}\sum_{r=1}^{R_l}\ell\left(\boldsymbol{s}_l^r\left(x_T'\right), \boldsymbol{s}_l^r(x_S), \boldsymbol{s}_l(x_T)\right). \tag{4}$$

3) Content Consistency Constraints: The constraint of content consistency primarily manifests in terms of semantics, image details, and textures, which is a significant challenge in RS image translation task. This study employs content contrastive loss, identity loss, and semantic loss to enforce content consistency between the input and translated images. For content contrastive loss, similar to style contrastive loss, we pass the L-layer feature maps of the encoder of $\boldsymbol{I}_{\text{CDIT}}$ through a content projection network $\boldsymbol{H}_C$ to get the projected content vectors, defined as follows:

$$\{\boldsymbol{c}_l(x)\}_L = \left\{\boldsymbol{H}_C\left(\boldsymbol{I}_{\text{CDIT}}^{\text{enc}-l}(x)\right)\right\}_L \tag{5}$$

where $\boldsymbol{H}_C$ is a two-layer MLP with 256 units in each layer. Additionally, we denote the specific location feature as $\boldsymbol{c}_l^s(x), s \in \{1, \ldots, S_l\}$ and the other features as $\boldsymbol{c}_l^{S\setminus s}(x)$, where S_l is the number of patches in each layer, set to 256.

To constrain the content consistency between the fake $x_T{}'$ and real x_T, we use (5) and match corresponding patches $\boldsymbol{c}_l^s(x_T{}')$ and $\boldsymbol{c}_l^s(x_T)$ at a specific location. We use the remaining patches in the real image x_T as negative examples, denoted as $\boldsymbol{c}_l^{S\setminus s}(x_T)$. The content contrastive loss is defined as follows:

$$\mathcal{L}_{\text{C}-\text{contra}} = \sum_{l=1}^{L}\sum_{s=1}^{S_l}\ell\left(\boldsymbol{c}_l^s(x_T{}'), \boldsymbol{c}_l^s(x_T), \boldsymbol{c}_l^{S\setminus s}(x_T)\right). \tag{6}$$

We utilize identity loss to preserve detail consistency between the real x_S and fake $x_S{}'$ generated by $\boldsymbol{I}_{\text{CDIT}}$

$$\mathcal{L}_{\text{id}} = \|x_S' - x_S\|_1. \tag{7}$$

Furthermore, we incorporate the cross-entropy loss $\mathcal{L}_{\text{seg}}$ of the semantic segmentation model as the supervision in the training of the image translation model.

4) Total Loss: We summarize the aforementioned losses to formulate the final objective function of our model. Specifically, we adopt adversarial loss functions from [46] to match the distribution of translated images T' and S' with the data distribution in the domain S, defined as $\mathcal{L}_{\text{adv}}$. To train the generator $\boldsymbol{I}_{\text{CDIT}}$ and the discriminator $\boldsymbol{D}$, the objective is to minimize the following two equations, respectively:

$$\mathcal{L}_G = \lambda_1\mathcal{L}_{\text{adv}_G} + \lambda_2\mathcal{L}_{\text{S}-\text{contra}} + \lambda_3\mathcal{L}_{\text{C}-\text{contra}} + \lambda_4\mathcal{L}_{\text{id}} + \lambda_5\mathcal{L}_{\text{seg}} \tag{8}$$

$$\mathcal{L}_D = \lambda_1\mathcal{L}_{\text{adv}_D} \tag{9}$$

where the hyperparameters λ_1–λ_5 control the balance among the different loss terms.

C. Stage B: CMG Based on Meta-Learning

1) Pseudo-Labels Generation: Pseudo-labeling is a self-training approach that generates pseudo-labels for unlabeled datasets and then uses them for joint training with labeled datasets. However, if pseudo-labels contain substantial inaccuracies, they can adversely degrade the model's training performance. In this article, our open compound target domain encompasses a diverse set of images with a wide range of features. Direct pseudo-labeling by a model pretrained on the source domain can lead to significant labeling inaccuracies due to domain shift. Therefore, in the first stage, we align the appearance of target images to the source domain. The translated images are then fed into the pretrained model to generate pseudo-labels. Additionally, to further avoid overfitting incorrect pseudo-labels and improve the effectiveness of meta-learning, we apply a filtering process to screen the pseudo-labels. Specifically, we leverage the entropy of each pixel's probability distribution [47] to screen high confidence pseudo-labels. First, we denote $p_{ij} \in \mathbb{R}^N$ as the softmax probabilities at pixel (i, j), where N is the number of classes, and its entropy is computed by the following equation:

$$\mathcal{H}\left(p_{ij}\right) = -\sum_{n=1}^{N} p_{ij}(n)\log p_{ij}(n) \tag{10}$$

where $p_{ij}(n)$ is the value of softmax probability at nth class. Then, we assess the reliability of the predictions and assign pseudo-labels using the following approach:

$$\hat{y}_{ij} = \begin{cases} \arg\max\limits_{n} p_{ij}(n), & \text{if } \mathcal{H}\left(p_{ij}\right) < \gamma \\ \text{ignore}, & \text{otherwise} \end{cases} \tag{11}$$

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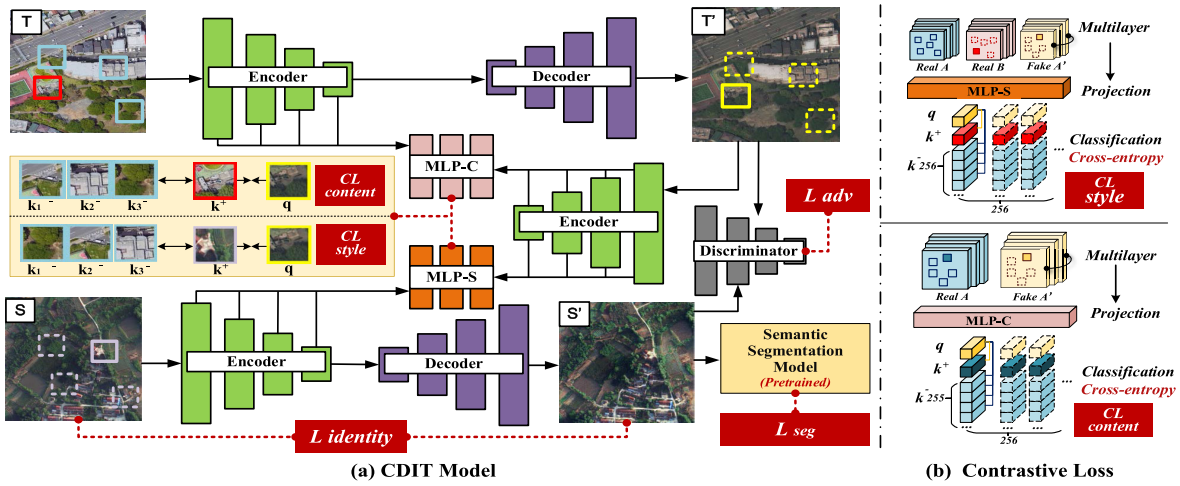


Fig. 3. (a) Schematic illustration of the proposed CDIT network. (b) Architecture of patchwise content and style contrastive losses.

where γ is the entropy threshold, dynamically set as $\gamma = np \cdot \text{percentile}(H \cdot \text{flatten}(), 70)$, with $\mathcal{H}$ representing the per-pixel entropy map of each image. Finally, we calculate the proportion of ignored pixels in each image. If the ignored pixels exceed 50%, the image is considered unreliable and its pseudo-labels are excluded from subsequent joint training.

2) DCWM Model: It aims to build an externally robust categorical memory matrix $\mathcal{M} \in \mathbb{R}^{N\times C}$ for each class, where N represents the number of classes and C denotes the feature channels. This matrix guides the semantic segmentation from different domains and is incorporated with the segmentation backbone network for joint training. Specifically, we design a memory update module ***Update*** and a memory read module ***Read*** to store the common features of each class into their respective memory matrix $\mathcal{M}$. The memory update and reading operations are illustrated in Fig. 4.

a) Memory update: During training, the module ***Update*** utilizes source-domain data with their labels to extract and update the categorical memory features $\widehat{\mathcal{M}} = \boldsymbol{Update}(\mathcal{M}, \boldsymbol{Enc}(x_S))$, where $\mathcal{M}$ represents the memory matrix from the previous iteration and ***Enc*** denotes the encoder of the semantic segmentation model. As depicted in Fig. 4(a), first, the input to the update network is the semantic feature map $\mathcal{F} = \boldsymbol{Enc}(x_S)$, where $\mathcal{F} \in \mathbb{R}^{H'\times W'\times C}$ and $x_s \in \mathbb{R}^{H\times W\times 3}$. Here, $H \times W$ is the original size of the source image x_s, and $H' \times W'$ is the feature size after convolutional and pooling operations of the pretrained encoder of the semantic segmentation model. Next, we feed $\mathcal{F}$ into a residual structure with a 1×1 convolutional layer to obtain $\mathcal{F}' \in \mathbb{R}^{C\times H'\times W'}$. Meanwhile, the ground-truth masks $\mathcal{Y}$ of source images guide the update of the nth item $\mathcal{M}(n)$ in the classwise memory. Specifically, $\mathcal{Y}$ is one-hot encoded and downsampled to a uniform size of $H' \times W'$ as $\mathcal{Y}' \in \mathbb{Z}^{N\times H'\times W'}$. We then calculate the temporary memory feature $\mathcal{T}$ by multiplying $\mathcal{F}$ with the mask of $\mathcal{Y}'$ classes as follows:

$$\mathcal{T}[n] = \frac{\mathcal{F}'\mathcal{Y}[n]^\top}{\text{sum}[n]} \tag{12}$$

where $\mathcal{T}[n]$ is a mean feature vector for each class and sum$[n]$ indicates the sum of non-zero pixels for each class. Finally, the updated memory feature $\widehat{\mathcal{M}}$ is obtained by adding $\mathcal{T}$ to $\mathcal{M}$ for each class, weighted by k as follows:

$$\widehat{\mathcal{M}}[n] = k \cdot T[n] + (1-k) \cdot M[n] \tag{13}$$

where k is set to 0.3 empirically.

During the initial training iteration $i = 0$, the memory matrix $\mathcal{M}$ is initialized as the mean semantic feature vector for each class, using ground-truth segmentation maps to mask the regions. The initial parameters of ***Enc*** are derived from a semantic segmentation model pretrained on the source domain.

b) Memory reading: As shown in Fig. 4(b), the module ***Read*** compensates the semantic features $\mathcal{F} \in \mathbb{R}^{H'\times W'\times C}$ extracted by the encoder of the semantic segmentation model, to obtain the memory-guided feature map $\widehat{\mathcal{F}} = \boldsymbol{Read}(\mathcal{M}, \boldsymbol{Enc}(x_S))$, where $\widehat{\mathcal{F}} \in \mathbb{R}^{H'\times W'\times C}$. Inspired by Kim et al. [15], we first aggregate the corresponding memory items for each class at each feature location and normalize them using the softmax function as (14), yielding a categorical memory weight $\mathcal{W} \in \mathbb{R}^{H'\times W'\times N}$ from the stored memory matrix $\mathcal{M} \in \mathbb{R}^{N\times C}$. To compute the class-aware memory weights $\mathcal{W}$, we first reshape the feature map into a 2-D tensor $\tilde{F} \in \mathbb{R}^{(H'W')\times C}$, where each row $\tilde{F}_l$ corresponds to a feature vector at spatial location $l = (i, j)$. Then, for each location l, the attention weight for the nth class memory vector $M[n] \in \mathbb{R}^C$ is computed using a softmax function over the similarity scores

$$\mathcal{W}_l[n] = \frac{\exp\left(\tilde{F}_l M[n]^\top\right)}{\sum_{n=1}^{N}\exp\left(\tilde{F}_l M[n]^\top\right)} \tag{14}$$

where $W_l[n]$ denotes the categorical weight of the nth class memory at location l and N is the number of semantic classes.

To further restore the memory features for each channel in the feature space, we compute the memory compensation feature $\mathcal{F}_m = \text{WM}$, where $\mathcal{F}_m \in \mathbb{R}^{H'\times W'\times C}$, by multiplying the weight $\mathcal{W}$ with the memory matrix $\mathcal{M}$. Finally, we concatenate the compensated memory feature $\mathcal{F}_m$ with the original semantic feature $\mathcal{F}$ and pass them through a 1×1 convolutional layer to obtain the final memory-guided feature $\widehat{\mathcal{F}}$. Herein, $\widehat{\mathcal{F}}$ is continuously fed to the decoder ***Dec*** of the semantic segmentation model for semantic label prediction.

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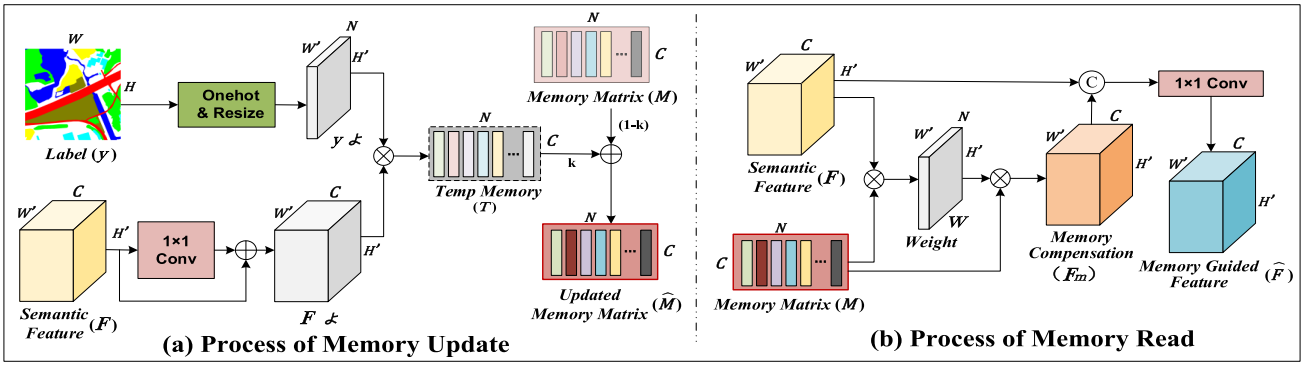


Fig. 4. Overview of the memory update and reading processes. (a) Memory update: source-domain labels guide categorical memory matrix updates. (b) Memory read: semantic features are compensated via memory weights computed between features and the matrix.

3) Categorical Memory Learning Strategy: The main challenge of this method is how to learn and memorize domain-agnostic categorical features from cross-domain datasets to improve pixel category discrimination. To this end, we employ a meta-learning framework that is jointly optimized alongside the semantic segmentation network. Each training iteration includes both a meta-training step and a meta-testing step, which are used to iteratively train the DCWM and simulate virtual testing, respectively. The detailed process of categorical memory learning is illustrated in Fig. 5 and Algorithm 1.

a) Optimization objective: Regarding the optimization objectives during the training process, we compute the pixelwise cross-entropy loss between the predicted images and the ground truth, defined as the semantic segmentation loss $\mathcal{L}_{\text{seg}}$ in the meta-training step. In contrast, during the meta-testing step, we utilize pseudo-labels from the target domain to compute the pixelwise cross-entropy loss $\mathcal{L}_{\text{p-seg}}$. Additionally, we introduce a novel intraclass cohesion loss $\mathcal{L}_{\text{coh}}$ and an interclass divergence loss $\mathcal{L}_{\text{div}}$ to enhance the categorical representation performance of the memory features. The intraclass cohesion loss encourages the aggregated semantic features, based on each memory item, to maintain local consistency with the ground truth. It calculates the distributional difference between the categorical memory weight $\mathcal{W}$ in the memory reading module and the ground truth as follows:

$$\mathcal{L}_{\text{coh}} = -\frac{1}{H'W'}\sum_{i=1}^{H'W'} \mathcal{Y}[i]\log(\mathcal{W}[i]). \tag{15}$$

The interclass divergence loss ensures that the features in the memory matrix for different classes are as distinct as possible, promoting a sufficient separation of classwise features for better discrimination. We introduce the cosine loss to compute the similarity between items of different categories in the newly updated memory matrix $\widehat{\mathcal{M}}$. $\mathcal{L}_{\text{div}}$ is formulated as follows:

$$\mathcal{L}_{\text{div}} = \sum_{n'\neq n}^{N} \frac{\max\left(\cos\left(\widehat{\mathcal{M}}[n]\widehat{\mathcal{M}}[n']\right), 0\right)}{N(N-1)}. \tag{16}$$

b) Meta training: The goal of meta-training is to learn the categorical memory features from the source domain and iteratively update the memory matrix. As shown in Fig. 5(a), memory updating and reading are performed simultaneously. Given a set of source-domain images x_S, the extracted semantic features are input into the memory reading module, which generates semantic features guided by the memory features $\mathcal{M}$ from the previous iteration. The memory-guided features are fed into the decoder of the semantic segmentation network to obtain the segmentation predictions. We then compute the $\mathcal{L}_{\text{seg}}$ and $\mathcal{L}_{\text{coh}}$ losses values to optimize the parameters of the encoder ***Enc***, decoder ***Dec***, and memory reading network ***Read***

$$\Theta'_{\text{Enc,Read,Dec}} \leftarrow \alpha\nabla_{\Theta}\left(\mu_1\mathcal{L}_{\text{seg}} + \mu_2\mathcal{L}_{\text{coh}}\right). \tag{17}$$

Simultaneously, we update the memory matrix $\widehat{\mathcal{M}}$ for this meta-training and optimize the parameters of the encoder ***Enc*** and memory update network ***Update*** with $\mathcal{L}_{\text{div}}$

$$\Theta'_{\text{Enc,Update}} \leftarrow \alpha\nabla_{\Theta}\mu_3\mathcal{L}_{\text{div}} \tag{18}$$

where α is the learning rate of the meta-training step. The first optimization aims to effectively guide the feature map using previously saved memory to train a stable semantic segmentation model. The second optimization focuses on saving even better patterns into memory and increasing the divergence between memory items.

c) Meta testing: Meta testing primarily optimizes the memory matrix and the semantic segmentation network to make them more adapted to the target and the unseen domains by testing their performance on cross-domain images, which is the most crucial step of our method. Specifically, as depicted in Fig. 5(b), we first refresh the memory $\mathcal{M}'$ with the parameters $\Theta'_{\text{Enc,Read,Dec,Update}}$ obtained during the training process, along with the source images x_S. This updated memory $\mathcal{M}'$ is then used to guide segmentation of the target images x_T. By computing $\mathcal{L}_{\text{p-seg}}$ and $\mathcal{L}_{\text{p-coh}}$, we optimize the parameters of the encoder ***Enc***, decoder ***Dec***, as well as memory guidance model ***Update*** and ***Read*** as follows:

$$\Theta^*_{\text{Enc,Update,Read,Dec}} \leftarrow \beta\nabla_{\Theta}\mu_4\mathcal{L}_{\text{p-seg}} + \mu_5\mathcal{L}_{\text{p-coh}} \tag{19}$$

where $\mathcal{L}_{\text{p-coh}}$ is calculated with the pseudo-labels of the target domain based on (15), and β is the learning rate of the meta-testing step, which is different from α. Notably, we freeze the encoder parameter Θ'_{Enc} in the memory reupdate step in this optimization, which can avoid unstable meta-learning caused by the asynchronous gradient updates between the encoder

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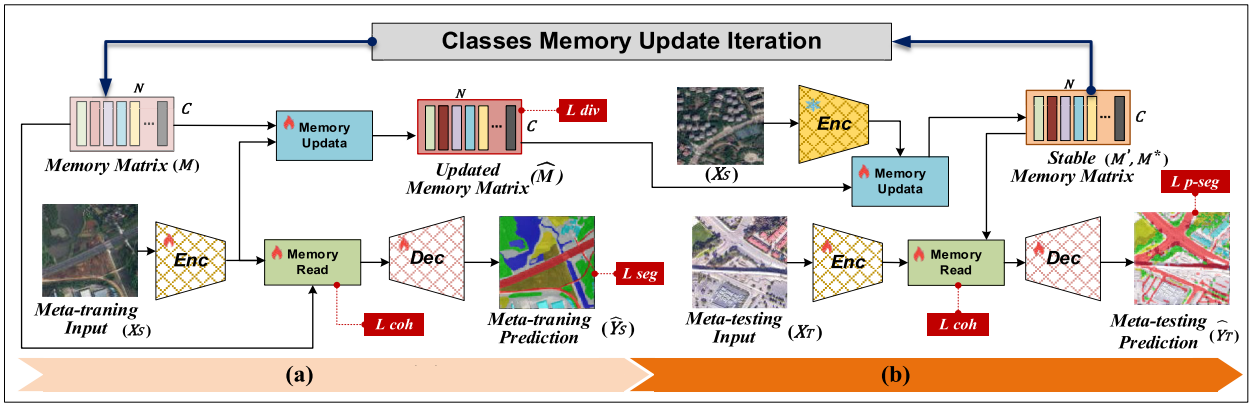


Fig. 5. Meta-learning process of the DCWM, including (a) meta-training and (b) meta-testing steps within every class memory update iteration.

Algorithm 1 Categorical Memory Learning Procedure

Input: Source domain image $x_s \in S$ with label y_s.
Target domain image $x_T \in T$ with pseudo-label $\hat{y}_T$.
Initialize: Memory matrix M at $i = 0$
Encoder of semantic segmentation Θ_{Enc}
Decoder of semantic segmentation Θ_{Dec}
Memory network Θ_{Update}, Θ_{Read}
While $i < I$ **do:**
Meta-training:
1: Sample batch $x_S = \{x_S^b\}_{b=1}^B$ from S.
2: Predict label $y'_S = Dec(Read(M, Enc(x_S)))$.
3: Compute $\mathcal{L}_{seg}$ with (y'_S, y_s).
4: Compute $\mathcal{L}_{coh}$ with $(M, y_s, Enc(x_S))$.
5: Update $\Theta'_{Enc,Read,Dec} \leftarrow \nabla_\Theta \lambda_1 L_{\text{seg}} + \lambda_2 L_{\text{coh}}$
6: Update memory $\widehat{\mathcal{M}} \leftarrow Update(M, Enc(x_S))$.
7: Compute $\mathcal{L}_{div}$ with $\widehat{\mathcal{M}}$.
8: Update $\Theta'_{Enc,Update} \leftarrow \nabla_\Theta \lambda_3 L_{\text{div}}$
Meta-training:
9: Sample batch $x_S = \{x_S^b\}_{b=1}^B$ from S.
10: Update memory $M' \leftarrow Update(\widehat{\mathcal{M}}, Enc'(x_S))$.
11: Sample batch $x_T = \{x_T^b\}_{b=1}^B$ from T.
12: Predict label $y'_T = Dec'(Read'(M, Enc'(x_T)))$.
13: Compute $\mathcal{L}_{p-seg}$ with $(y'_T, \hat{y}_T)$.
14: Compute $\mathcal{L}_{coh}$ with $(M, \hat{y}_T, Enc'(x_T))$.
15: Update $\Theta^*_{Enc,Update,Read,Dec} \leftarrow \nabla_\Theta \lambda_1 L_{\text{p-seg}} + \lambda_2 L_{\text{coh}}$.
16: Update memory $M^* \leftarrow Update^*(\widehat{\mathcal{M}}, Enc^*(x_S))$.
$\Theta_{Enc,update,read,dec} \leftarrow \Theta^*_{Enc,update,read,dec}$
$M \leftarrow M^*$
$i \leftarrow i + 1$

and the other networks. Thus, this novel step also enables the parameters Θ_{Update} to receive second-order gradient feedback on whether the updated memory $\mathcal{M}'$ is adaptive across different domains. Finally, we update the memory $\mathcal{M}^*$ with the optimized parameters $\Theta^*_{\text{Enc,Update}}$ to serve as the initial memory matrix for the next iteration.

Overall, the meta testing step facilitates the abstraction of domain-agnostic categorical features from the source domain into the current memory matrix, ensuring the generalization of the memorized features to the target and unseen domains. Furthermore, the inference of the entire framework is depicted in Fig. 2. We apply the well-trained parameters of both the memory module and the semantic segmentation network, obtained through the meta-learning, to perform image prediction.

TABLE I
DIVISIONS OF THE EXPERIMENTAL DATASETS

Datasets	Resolution	Number	Usage
Loveda (S)	0.3m	4190 (512×512)	Source
Inria (A)	0.3m	1484(512×512)	Target
Open Cities AI(A)	0.2m	1660(768×768)	Target
GeoNRW(A)	0.3m	1176(512×512)	Target
LandCover.ai (A)	0.3-0.5 m	273(1024×1024)	Unseen
xBD (S)	0.5 m	455(1024×1024)	Unseen
Mixed Set (A/S)	0.3m	1377(512×512)	Unseen

* S: Spaceborne; A: Airborne

IV. Experimental Results

In this section, we evaluate our approach for OCDA semantic segmentation, detailing the experimental setup, comparing it quantitatively and qualitatively with the state-of-the-art UDA methods in RS, and conducting ablation studies to assess key components.

A. Experimental Setup

1) Datasets: This study focuses on domain adaptation for multiclass semantic segmentation in very high-resolution visible imagery. Existing RS datasets often exhibit inconsistencies in category space and variations in annotation precision. The OpenEarthMap dataset [48] unifies label spaces across multiple subdatasets with high accuracy by reusing benchmark imagery and manually annotating new land cover labels. This provides a solid foundation for cross-domain adaptation.

For domain configuration, we use subsets from the OpenEarthMap [48] along with the LoveDA dataset [49] to conduct our experiments. A summary of the datasets is presented in Table I, while the experimental domain setup is illustrated in Fig. 6. The detailed descriptions are as follows.

LoveDA [49] consists of 5987 submeter satellite images from rural and urban areas in China, with 4190 annotated images. We use it as the source domain.

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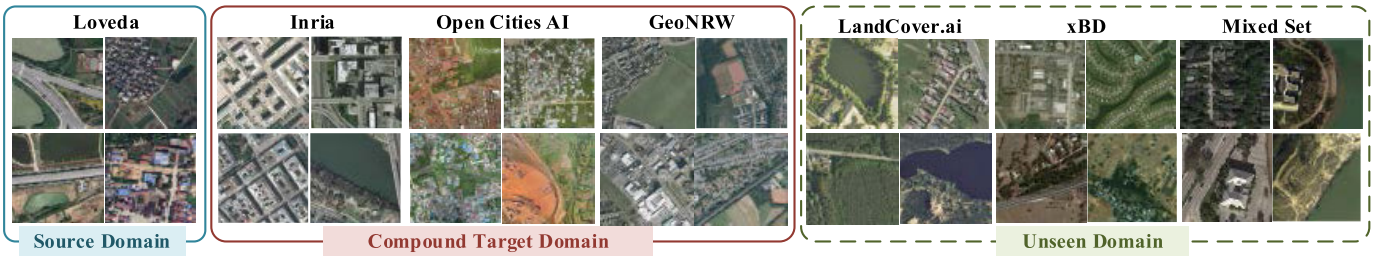


Fig. 6. Examples of datasets used in OCDA experiments.

Inria [50] was originally a binary building segmentation dataset with aerial imagery from six global regions. We incorporate its relabeled version [48] into the compound target domain.

Open Cities AI [51] provides 0.2-m resolution aerial images from ten African regions, primarily for building segmentation. We incorporate the relabeled subset [48] into the compound target domain.

GeoNRW [52] includes aerial images, LiDAR-derived digital elevation models, and segmentation maps with ten land cover classes from Germany. We use its relabeled subset [48] as part of the compound target domain.

LandCover.ai [53] focuses on automatic land cover mapping (buildings, woodlands, water, and roads) from aerial imagery in Poland. We adopt its relabeled subset [48] as an unseen domain for aerial imagery adaptation.

xBD [54] is a large-scale high-resolution satellite dataset for building damage detection. We utilize its relabeled subset [48] as an unseen domain for satellite imagery adaptation.

Mixed Set comprises high-resolution RS images from Google Earth and aerial cameras. We use it exclusively for qualitative assessments to evaluate the model's generalization.

Buildings, water, low vegetation, forests, barren areas, roads, and background are selected as the standard categories for classification across these datasets. To align with the classification benchmark, "rangeland," "agriculture land" from OpenEarthMap, and "agriculture" from LoveDA are mapped to "low vegetation," while "developed space" is classified as "background." To the best of our knowledge, this is the first OCDA experiment using RS imagery from aerial and satellite sources. We cropped image patches from different datasets for model training following (1).

2) Implementation Details: By default, all input images are resized to a uniform patch size of 512 × 512 for training. We employ DeepLabV3+ [45] as the semantic segmentation architecture, pretrained it on the source-domain dataset. More specifically, the image translation model CDIT is trained with a batch size of 2 for 12 epochs using the Adam optimizer with an initial learning rate of 2×10^{-4}, λ_1, λ_3, and λ_4 are empirically set to 1, while λ_2 and λ_5 are set to 0.5. For the DCWM model, the memory dimension is set to 512. We jointly train DCWM and the semantic segmentation model through iterative meta-training and meta-testing with a batch size of 8. The inner learning rate for meta-training is set to 10^{-3}, while the outer learning rate for meta-testing is set to 10^{-2}. The hyperparameters of the loss function are set as follows: $\mu_1 = 1$, $\mu_2 = 0.02$, $\mu_3 = 0.4$, $\mu_4 = 0.5$, and $\mu_5 = 0.02$. To compare with the proposed method, we reimplemented several domain adaptation approaches. All experiments are conducted on a Nvidia Gefore GTX TITAN GPU. Regarding the time cost of the two-stage training, the CDIT network achieved favorable performance after 12 epochs on the experimental dataset using a single GPU, with a total training time of approximately 5 h. The second-stage model required 200 epochs, with a training time of approximately 23 h on a single GPU.

3) Evaluation Protocol:

a) Image translation: To evaluate the performance of the CDIT network in image translation, we employ four image quality assessment metrics commonly used in image reconstruction: kernel inception distance (KID) [55], Fréchet inception distance (FID) [56], inception score (IS) [57], and natural image quality evaluator (NIQE) [58]. Among them, KID and FID can effectively evaluate the perceptual distance between the translated images and the target dataset. IS and NIQE can assess image quality in the absence of reference images.

b) Semantic segmentation: For the multiclass segmentation task, we utilize the classic mean intersection over union (mIoU) to evaluate the classification results at the class level, alongside overall accuracy (OA) to assess the aggregate errors across all classes. The mIoU and OA are defined as follows:

$$\text{mIoU} = \frac{1}{C}\sum_{i=1}^{c}\frac{\text{TP}_C}{\text{TP}_C + \text{FP}_C + \text{FN}_C} \tag{20}$$

$$\text{OA} = \frac{\sum_{i=1}^{C}\text{TP}_C}{\sum_{i=1}^{C}(\text{TP}_C + \text{FN}_C)} \tag{21}$$

where TP, FP, and FN represent the true positives, false positives, and false negatives, respectively.

B. Comparison With the State of the Art

1) Results on Image Translation: To assess the practical effectiveness of the proposed CDIT network, we conduct a parallel comparison with several classical and state-of-the-art style transfer methods, including MKL [27], a color alignment method for images based on the Monge-Kantorovich theory; WCT2 [35], a photorealistic style transfer method that combines wavelet convolution and WCT; CAST [33], a contrastive learning-based style transfer method; and RSIT [59], a variant of CycleGAN dedicated to RS image translation.

Fig. 7 qualitatively illustrates the translation results of our proposed method for two different images. We observe that our method achieves a robust mapping of intraclass semantics

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TABLE II
QUANTITATIVE EVALUATION ON IMAGE TRANSLATION, MODEL PARAMETERS, AND COMPUTATIONAL COMPLEXITY

Methods	Target Domains → LoveDA				Params (M)↓	FLOPs (G)↓
	KID↓	FID↓	IS↑	NIQE↓		
No Trans	5.875	1.061	1.324	**11.920**	—	—
MKL [27]	5.850	1.023	1.313	13.487	—	—
WCT2 [35]	6.082	1.021	**1.337**	14.479	10.12	130.74
CAST [33]	7.495	1.217	1.320	18.917	**3.51**	146.78
RSIT [59]	6.724	1.067	1.296	15.975	12.63	216.95
CDIT(Ours)	**4.637**	**0.884**	1.325	14.644	8.38	**122.42**

before and after image translation (e.g., water and buildings), rather than merely performing color matching. Compared to MKL, WCT2, and RSIT, the proposed method achieves more realistic translation and better alignment with the style dataset, exhibiting strong consistency in both image texture details and semantic accuracy. The images generated by the CAST method are more abstract, with significant losses in content details. Furthermore, the quantitative comparison with relevant methods is reported in Table II. CDIT outperforms other methods in terms of FID and KID scores, which proves that the translated image distribution closely aligns with that of real datasets. In terms of image quality evaluation, although MKL demonstrates the best NIQE score and WCT2 achieves the highest IS score, these results emphasize the advantages of traditional methods in maintaining image sharpness and details. Nonetheless, these approaches rely heavily on the manual selection of a highly correlated single image as a style reference, and they primarily focus on color matching while neglecting the issue of semantic consistency. Compared to deep learning methods for style transfer, such as CAST and RSIT, our approach achieves superior image quality metrics.

Table II also provides a comparison of the number of parameters (M) and computational complexity in terms of FLOPs (G). To ensure a fair comparison, we calculated all models, using an input size of 500 × 500. Notably, our CDIT has the second fewest parameters among DL-based methods and achieves the lowest computational complexity in terms of FLOPs compared to other baselines, highlighting its efficiency. Therefore, our CDIT model can serve as a flexible and practical approach, particularly for tasks that require joint training of image translation models with other components.

2) Results on Cross-Domain Segmentation: To demonstrate the effectiveness of our framework, we compare it with five other state-of-the-art domain adaptation approaches with different strategies. AdaptSeg [18] is a classic GAN-based UDA method that employs feature space alignment. ResiDualGAN [32] is a UDA method specifically designed for RS images, integrating image translation with output space adaptation. LTIR [31] is a two-stage UDA method that leverages image style transfer to enrich sample diversity and combined with pseudo-label self-training. ADVENT [28] uses entropy loss to minimize entropy and adversarial loss indirectly to reduce the gap between the source and target domains. OCDA [9] is an OCDA method that primarily utilizes curriculum learning and a memory bank to guide the generalization of the model to unseen domains. We also list the baseline results of the model trained solely on the source domain, denoted as "Source_Only."

A visual comparison of the segmentation results on different datasets is shown in Fig. 8. Under the same experimental settings, our method demonstrates robust adaptation capabilities. Specifically, in the regions indicated by green arrows, our method demonstrates superior ability in distinguishing object boundaries (e.g., roads and buildings) and accurately segmenting fine-scale structures, such as small tree patches. This advantage is particularly evident on unseen datasets, where the model successfully identifies most land-cover categories despite the domain shifts. Table III lists the mIoU and OA metrics of different domain adaptation methods across the source, compound target, and unseen domains. Due to the differences in label precision between the LoveDA labels and the ground truth of other datasets, we observe that all methods exhibit lower scores on the OpenEarthMap subsets. Nonetheless, our method achieves effective performance across both the compound target and unseen domains. Specifically, for the source domain, our method preserves the source classification knowledge during the adaptation to the target domain, which is in marked contrast to other methods that focus only on adaptation to the target domain with a catastrophic forgetting in the source domain. For the compound target datasets, our method outperforms the ResiDualGAN approach, as well as the multitarget-domain adaptation methods ADVENT and LTIR. For unseen domain adaptation, the classical UDA settings struggle to generalize to unseen domain images, while the OCDA methods exhibit significant performance gains. Among them, OCDA ranks second only to our proposed method. To further exhibit the generalization of our model to unseen domains, we conducted experiments on large-scale real-world aerial RS images in Mixed Set. The semantic segmentation results of large-scale images, along with zoomed-in detail, are presented in Fig. 9. These tests highlight the robustness and applicability of our approach in real-world RS tasks.

Furthermore, to compare the performance of different UDA methods in discriminating seven land-cover categories, Fig. 10 presents a t-SNE visualization comparing the embeddings obtained from various approaches. These include the baseline method without domain adaptation, the suboptimal method ResiDualGAN on the compound target domain, the suboptimal method OCDA on the unseen domain, and our two methods: the first-stage image alignment method "CDIT" and the entire domain adaptation framework "CDIT + DCWM." We randomly sample embeddings from the compound target and unseen datasets for visualization. Compared to other methods, our approach produces more class-discriminative features, although some ambiguity remains for certain categories. Fig. 10 also reveals that buildings, low vegetation, and roads are challenging to distinguish, with significant confusion observed across two domains.

As RS images are influenced by diverse sensor platforms, leading to substantial style discrepancies, UDA models usually exhibit poor performance, especially in cross-platform and cross-resolution conditions. Since this study focuses on very high-resolution RS images, we mitigate the effect of resolution and sensor platform differences on the domain adaptation model to some extent by preprocessing for image

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Fig. 7. Image translation results. Style alignment of compound target-domain images to the LoveDA dataset.

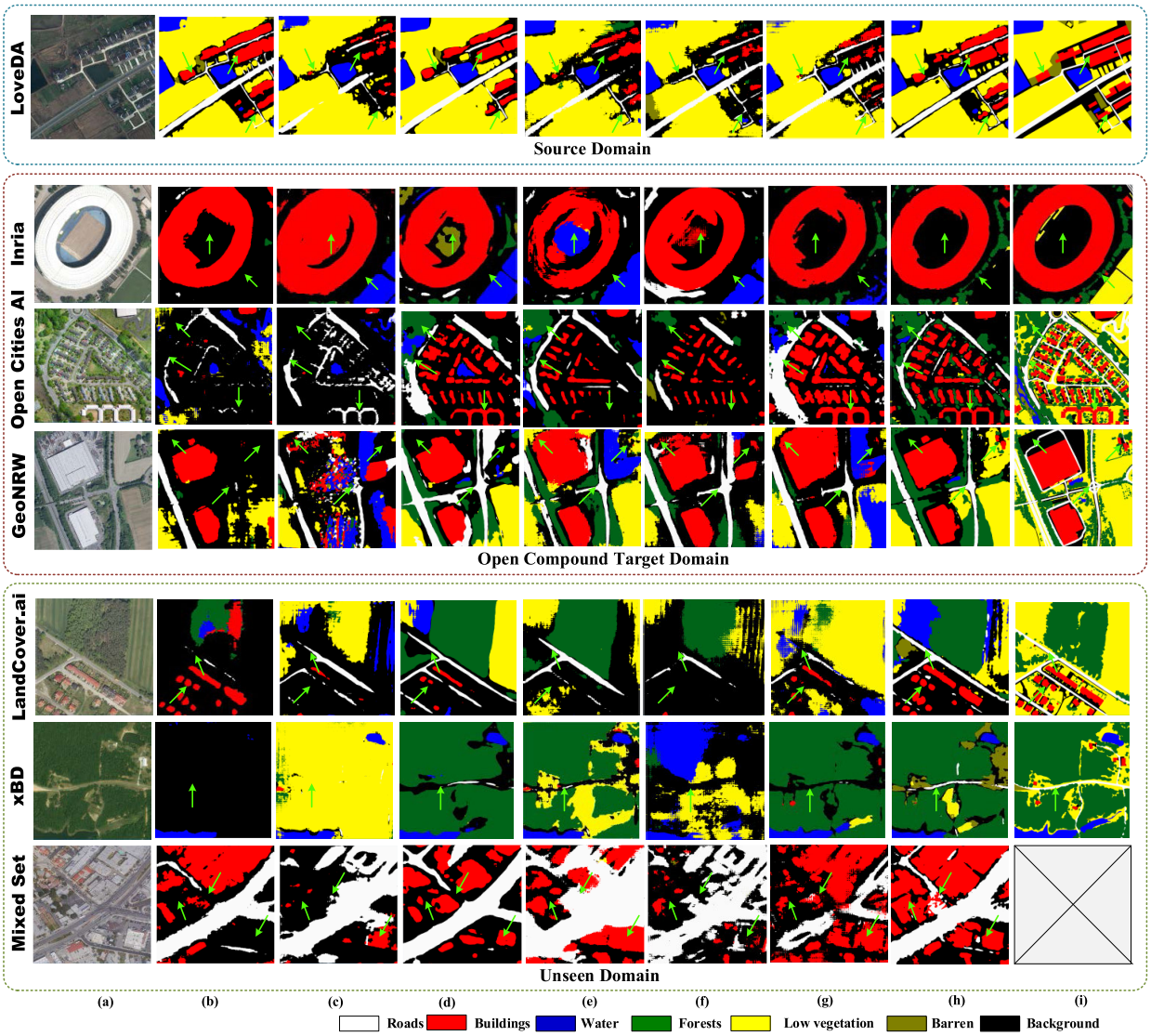


Fig. 8. Segmentation results of different methods on the source, target, and unseen domains. All methods are trained under the same OCDA setting. Green arrows highlight representative regions, where our method demonstrates superior performance(a) Image. (b) Source_only. (c) Adaptseg. (d) ResiDualGAN. (e) LTIR. (f) ADVENT. (g) OCDA. (h) Ours. (i). GT.

scale adjustment. As shown in Table III, the domain adaptation yields notable improvements on both xBD dataset (Spaceborne) and LandCover.ai dataset (Airborne). Qualitative segmentation results further validate this, particularly for the Mixed Set, which combines both satellite and aerial imagery.

Table IV compares the computational cost of our method with existing UDA approaches. For a fair evaluation, we compute all models using an input size of 512 × 512. Since our method requires memory mechanism, the number of parameters increases slightly. However, computational complexity

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TABLE III
CROSS-DOMAIN RS IMAGE SEMANTIC SEGMENTATION COMPARISON RESULTS: FROM LOVEDA TO OPEN COMPOUND TARGET DOMAINS AND UNSEEN DOMAINS

Methods	Source		Compound target						Unseen			
	LoveDA		Inria		Open Cities AI		GeoNRW		LandCover.ai		xBD	
	mIoU↑	OA↑	mIoU↑	OA↑	mIoU↑	OA↑	mIoU↑	OA↑	mIoU↑	OA↑	mIoU↑	OA↑
Source-Only	**66.9**	**82.0**	30.2	37.1	23.9	35.4	16.8	32.4	20.6	39.7	14.2	31.2
AdaptSeg [18]	53.6	80.6	20.5	41.4	23.8	38.7	15.4	40.1	23.8	39.3	21.8	49.6
ResiDualGAN [32]	43.9	65.4	30.6	53.3	30.9	48.3	24.7	51.9	24.1	56.4	24.2	47.5
LTIR [31]	44.1	67.3	27.5	48.3	29.2	41.2	20.9	45.5	23.0	50.4	12.5	30.9
ADVENT [28]	46.7	67.8	23.2	38.9	28.2	34.7	19.9	42.3	21.6	37.6	13.4	34.7
OCDA [9]	53.3	79.9	28.3	50.8	27.6	39.9	22.0	47.8	26.9	40.5	25.4	50.8
Ours	61.3	80.9	**35.9**	**56.1**	**32.4**	**50.7**	**26.1**	**58.1**	**29.1**	**58.8**	**30.7**	**53.1**

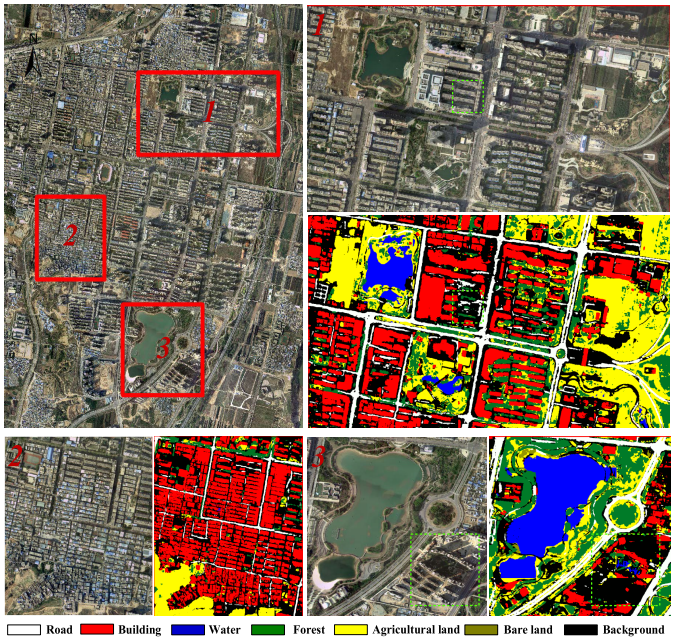


Fig. 9. Semantic segmentation results on unseen large-scale RS images, with zoomed-in detail.

in terms of FLOPs (G) is competitive with other methods. Our UDA model demonstrates strong generalization capability and effectively reduces computational overhead. It effectively balances model complexity and performance, ensuring both efficiency and accuracy. The reduced FLOPs enable faster inference and lower resource consumption, making our approach highly suitable for large-scale RS applications.

3) Discussion on Failure Results: Although the method in this article achieves notable accuracy improvements, its performance is sensitive to the quality of the pseudo-labels. Inaccurate pseudo-labels can adversely impact the overall effectiveness of domain adaptation. In addition, segmentation errors are more likely to occur in large, contiguous land-cover areas with weakly textured feature (e.g., low vegetation and water bodies). For unseen DG, our method can only improve the segmentation results to a certain extent. Even so, this method offers an iterative update mechanism that allows for the continuous integration of unknown domains into the compound target domain, thereby improving the model's generalization over time.

C. Ablation Study

In this section, we conduct ablation experiments to verify the effectiveness of key components and strategies of our framework in two major parts.

1) Study of Image Translation Model: To verify the contribution of each component in the CDIT network, we evaluate the effect of the contrastive learning temperature parameter τ on the results, as well as the influence of content contrastive loss and style contrastive loss on the training process. Additionally, we analyze the performance gain of the entire image translation stage for cross-domain semantic segmentation tasks.

a) Influence of temperature parameter τ: To investigate the impact of the contrastive learning temperature parameter τ on model performance, we conducted a series of experiments by varying $\tau \in \{0.05, 0.07, 0.1, 0.5, 1.0\}$ within a predefined range while keeping all other settings unchanged. The qualitative results of the best epoch's image translation are summarized in Table V. Additionally, we present visualizations under different values of τ in Fig. 11. Experimental results indicate that an excessively small τ degrades the model's ability to extract semantic features during training. In contrast, a larger τ reduces the discriminative power of contrastive learning for negative samples, weakening feature separation between different classes. The optimal balance is observed at $\tau = 0.07$, where the model achieves the highest accuracy while maintaining stable convergence.

b) Influence of two contrastive losses: To evaluate the impact of content contrastive loss $\mathcal{L}_{\text{C-contra}}$ and style contrastive loss $\mathcal{L}_{\text{S-contra}}$ on model performance, we conducted ablation experiments by selectively removing each loss term and comparing the results with the full model. Table VI presents the quantitative comparison in terms of KID, IS, and the number of epochs required for optimal convergence. The results indicate that both content and style contrastive losses play crucial roles in improving model performance. Higher KID and lower IS scores indicate weaker style alignment and degraded image quality. Without content and style contrastive losses, relying solely on adversarial loss results in the worst performance and slowest convergence, requiring 30 iterations for stability. Removing $\mathcal{L}_{\text{C-contra}}$ increases FID and lowers IS, reducing cross-domain content preservation, while omitting $\mathcal{L}_{\text{S-contra}}$ results in a performance drop and requires more training epochs to align the dataset's style appearance. The full

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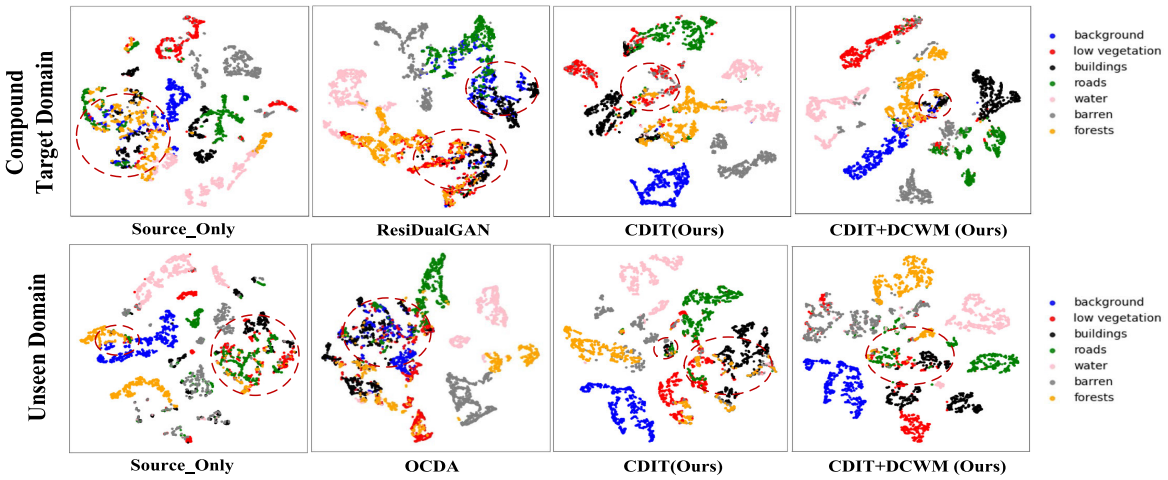


Fig. 10. t-SNE visualization of the obtained embeddings. Red circles indicate regions with easily confused embeddings.

TABLE IV

COMPARISON OF COMPUTATIONAL COST. TESTED WITH THE IMAGE SIZE OF 512 × 512

Methods	FLOPs (G)↓	Params (M)↓
AdaptSeg [18]	173.81	43.27
ResiDualGAN [32]	109.42	**26.01**
LTIR [31]	173.09	45.79
ADVENT [28]	175.34	43.27
OCDA [9]	184.96	46.04
Ours	**69.61**	45.37

TABLE V

QUANTITATIVE COMPARISON OF TRAINING RESULTS WITH DIFFERENT τ VALUES

τ	0.05	0.07	0.1	0.5	1.0
KID↓	4.98	**4.64**	4.82	5.79	7.94
FID↓	1.95	**0.88**	0.92	1.06	1.21

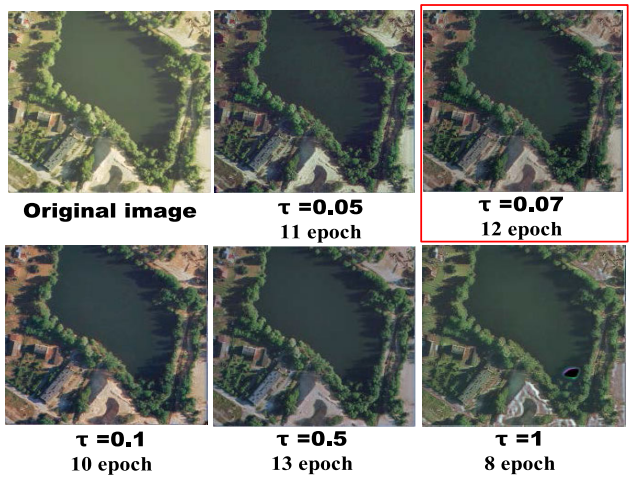


Fig. 11. Visualization of contrastive learning with varying temperature parameter τ.

model balances performance and efficiency, achieving optimal convergence in 12 epochs. These results confirm that content and style contrastive learning enhances detail preservation and rapid style adaptation, supporting the second-stage training.

c) Influence of image translation: To evaluate the first stage's effectiveness, we conduct cross-domain segmentation using CDIT alone ("Only CDIT"), with results presented in Table VII. Compared to the source-trained model ("Baseline"), both mIoU and OA improve significantly on target and unseen domains, highlighting the benefits of image-level alignment. Additionally, we compare the full framework ("full model") with a variant excluding this stage ("w/o CDIT"). Without image translation, the second stage relies solely on meta-learning for categorical feature memory. In contrast, incorporating CDIT enhances pseudo-label quality, enabling better adaptation to the target domain and improved generalization to unseen domains.

2) Study of Meta-Learning-Based Domain Adaptation Model: This part evaluates the contributions of each component in the second stage, including the pseudo-labeling strategy, the influence of memory guidance, and the impact of the two proposed losses ($\mathcal{L}_{\text{coh}}$ and $\mathcal{L}_{\text{div}}$) on the training process. For these experiments, we present evaluation results across three datasets from different domains: LoveDA, Open Cities AI, and xBD.

TABLE VI

IMPACT OF CONTENT AND STYLE CONTRASTIVE LOSSES

Model Variant	$\mathcal{L}_{\text{C}}$	$\mathcal{L}_{\text{S}}$	KID↓	IS↑	Convergence Epoch ↓
Full Model	✓	✓	**4.637**	**1.325**	**12**
w/o $\mathcal{L}_{\text{C-contra}}$	×	✓	7.551	1.107	25
w/o $\mathcal{L}_{\text{S-contra}}$	✓	×	5.568	1.243	17
w/o Both losses	×	×	8.699	1.056	30

a) Influence of pseudo-labeling strategy: To further investigate the effectiveness of the pseudo-label filtering strategy (PLS) on domain adaptation, we compare the model with pseudo-labeling strategy ("full model" in Table VII) against the model without it ("w/o PLS" in Table VII). Overall, the results in Table VII suggest that PLS enhances the model's adaptability to new domains to some extent. The accuracy of pseudo-labels significantly affects the generalization of the DA model to unseen domains. Furthermore, we examine the selection of the entropy threshold γ by testing different values

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TABLE VII
QUANTITATIVE COMPARISON OF DIFFERENT KEY COMPONENTS ACROSS THREE DATASETS UNDER VARIOUS TRAINING SETTINGS

Setting	CDIT	PLS	CMG	LoveDA (S)		Open Cities AI (T)		xBD (U)	
				mIoU↑	OA↑	mIoU↑	OA↑	mIoU↑	OA↑
Baseline	×	×	×	**66.9**	**82.0**	23.9	35.4	14.2	31.2
Only CDIT	✓	×	×	66.9	82.0	31.5	49.3	26.9	48.2
w/o CDIT	×	✓	✓	58.0	75.3	25.9	36.7	19.9	30.8
w/o PLS	✓	×	✓	60.3	78.6	31.4	47.5	28.7	50.1
w/o CMG	✓	✓	×	59.4	76.9	28.8	42.2	24.8	46.9
Full Model	✓	✓	✓	61.3	80.9	**32.4**	**50.7**	**30.7**	**53.1**

TABLE VIII
EVALUATION OF ENTROPY THRESHOLD γ IN PSEUDO-LABEL FILTERING

γ	LoveDA (S)		Open Cities AI (T)		xBD (U)	
	mIoU↑	OA↑	mIoU↑	mIoU↑	OA↑	mIoU↑
50%	60.9	77.2	29.7	45.2	25.4	46.3
70%	**61.3**	**80.9**	**32.4**	**50.7**	**30.7**	**53.1**
90%	61.0	79.2	31.9	48.3	29.6	52.2
100%	60.3	78.6	31.4	47.5	28.7	50.1

TABLE IX
EVALUATION OF MEMORY UPDATE WEIGHTING FACTOR k

k	LoveDA (S)		Open Cities AI (T)		xBD (U)	
	mIoU↑	OA↑	mIoU↑	mIoU↑	OA↑	mIoU↑
0.1	59.2	75.4	30.7	47.1	27.4	50.3
0.3	**61.3**	**80.9**	**32.4**	**50.7**	30.7	**53.1**
0.5	60.8	78.8	32.1	49.7	**30.8**	52.9
0.7	60.3	79.0	31.9	48.9	29.7	49.2
0.9	58.2	77.2	30.5	46.8	26.5	48.5

(50%, 70%, 90%, and 100%), with the corresponding mIoU and OA results reported in Table VIII. As shown, a lower γ (e.g., 50%) ignores too many pixels, significantly reducing the amount of usable training data and hindering adaptation. Conversely, a higher γ (e.g., 100%) retains excessive noisy pseudo-labels, leading to error accumulation and degraded segmentation performance. The best tradeoff is observed at $\gamma = 70$, which achieves the highest mIoU and OA while maintaining a balance between filtering noisy labels and preserving sufficient supervision.

b) Influence of meta-learning-based memory guidance model: Through the second-stage meta-learning framework, we obtain a robust class memory matrix. To assess the effectiveness of CMG, we conduct two ablation studies. First, we train a model without CMG, relying solely on pseudo-labels generated by CDIT to fine-tune a new semantic segmentation model ("w/o CMG"). The absence of CMG leads to a noticeable performance drop in both mIoU and OA, particularly in the unseen domain. This demonstrates that relying solely on pseudo-labels without a structured memory guidance limits the model's ability to generalize effectively. Next, we compare the full framework "full model" in Table VII, showing that CMG enhances adaptation by leveraging accumulated domain-invariant knowledge for more stable and accurate predictions. Additionally, we assess the second-stage-only model ("w/o CDIT") and the first-stage-only model ("CDIT"). While the first stage aligns images across domains but lacks strong semantic guidance, the second stage benefits from meta-learning yet struggles with noisy pseudo-labels. The full model, integrating both stages, achieves the best balance, demonstrating the complementary roles of CDIT and CMG in robust domain adaptation. Furthermore, to evaluate the impact of the memory update parameter k, we conduct a sensitivity analysis by varying k in the range $\{0.1, 0.3, 0.5, 0.7, 0.9\}$, while keeping the training duration fixed at 200 epochs for all models. As shown in Table IX, setting $k = 0.3$ achieves the best performance, indicating an optimal tradeoff between memory stability and adaptability to new domains.

TABLE X
INFLUENCE OF $\mathcal{L}_{\text{coh}}$ AND $\mathcal{L}_{\text{div}}$ ON MODEL PERFORMANCE

Model	$\mathcal{L}_{\text{coh}}$	$\mathcal{L}_{\text{div}}$	LoveDA (S)		Open Cities AI (T)		xBD (U)	
			mIoU↑	OA↑	mIoU↑	OA↑	mIoU↑	OA↑
Full	✓	✓	**61.3**	**80.9**	**32.4**	**50.7**	**30.7**	**53.1**
w/o $\mathcal{L}_{\text{coh}}$	×	✓	60.8	80.0	30.8	48.4	29.5	52.9
w/o $\mathcal{L}_{\text{div}}$	✓	×	60.5	79.7	29.2	46.2	28.4	51.3
w/o losses	×	×	59.7	78.8	28.6	45.6	26.5	49.1

c) Influence of two proposed losses: To verify the effectiveness of the two losses proposed in the second stage, we examine different combinations of intraclass cohesion loss and interclass divergence loss by testing different model variants with and without $\mathcal{L}_{\text{coh}}$ and $\mathcal{L}_{\text{div}}$ in Table X. We observe that $\mathcal{L}_{\text{div}}$ provides a higher gain in model adaptability, which we attribute to its ability to encourage greater independence among the domain-invariant memory features for each class, thereby facilitating better pixel classification. Similarly, incorporating $\mathcal{L}_{\text{coh}}$ alone improves mIoU and OA, highlighting its ability to enhance intraclass consistency. However, the best performance is achieved when both losses are combined, with the full model attaining the highest scores. Overall, these losses complement each other, jointly contributing to significant performance gains across datasets.

V. CONCLUSION

In this article, we propose a novel two-stage OCDA method for semantic segmentation in RS imagery. The first stage employs a CDIT model to reduce the appearance gap

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between compound target and source domains, producing reliable pseudo-labels. In the second stage, a DCWM model progressively updates abstract categorical features within a meta-learning framework, providing external class guidance for segmentation. Unlike traditional single-domain or multitarget adaptations with explicit boundaries, our method models the target domain as multiple homogeneous domains without clear boundaries, enhancing sample diversity. By combining image-level alignment and category-generalized memory features, our approach significantly improves adaptability to the target domain and generalization to unseen domains. Extensive experiments demonstrate the state-of-the-art performance on benchmarks and robust generalization to unseen RS images. With its flexibility and scalability, this method opens avenues for future research, including open-set domain adaptation.

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REFERENCES

[1] S. Ghaffarian, D. Roy, T. Filatova, and N. Kerle, "Agent-based modelling of post-disaster recovery with remote sensing data," *Int. J. Disaster Risk Reduction*, vol. 60, Jun. 2021, Art. no. 102285.

[2] B. Chen et al., "Hierarchical GNN framework for Earth's surface anomaly detection in single satellite imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5627314.

[3] D. Wittich and F. Rottensteiner, "Appearance based deep domain adaptation for the classification of aerial images," *ISPRS J. Photogramm. Remote Sens.*, vol. 180, pp. 82–102, Oct. 2021.

[4] O. Tasar, Y. Tarabalka, A. Giros, P. Alliez, and S. Clerc, "StandardGAN: Multi-source domain adaptation for semantic segmentation of very high resolution satellite images by data standardization," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 747–756.

[5] O. Tasar, S. L. Happy, Y. Tarabalka, and P. Alliez, "ColorMapGAN: Unsupervised domain adaptation for semantic segmentation using color mapping generative adversarial networks," *IEEE Trans. Geosci. Remote Sens.*, vol. 58, no. 10, pp. 7178–7193, Oct. 2020.

[6] J. He, X. Jia, S. Chen, and J. Liu, "Multi-source domain adaptation with collaborative learning for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 11003–11012.

[7] T. Isobe et al., "Multi-target domain adaptation with collaborative consistency learning," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8183–8192.

[8] K. Saito, K. Watanabe, Y. Ushiku, and T. Harada, "Maximum classifier discrepancy for unsupervised domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 3723–3732.

[9] Z. Liu et al., "Open compound domain adaptation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12403–12412.

[10] R. Gong et al., "Cluster, split, fuse, and update: Meta-learning for open compound domain adaptive semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 8340–8350.

[11] A. Gupta, J. Johnson, A. Alahi, and L. Fei-Fei, "Characterizing and improving stability in neural style transfer," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 4087–4096.

[12] E. Risser, P. Wilmot, and C. Barnes, "Stable and controllable neural texture synthesis and style transfer using histogram losses," 2017, *arXiv:1701.08893*.

[13] T. Park, A. A. Efros, R. Zhang, and J.-Y. Zhu, "Contrastive learning for unpaired image-to-image translation," in *Proc. Eur. Conf. Comput. Vis.*, 2020, pp. 319–345.

[14] J. R. Binder and R. H. Desai, "The neurobiology of semantic memory," *Trends Cogn. Sci.*, vol. 15, no. 11, pp. 527–536, Nov. 2011.

[15] J. Kim, J. Lee, J. Park, D. Min, and K. Sohn, "Pin the memory: Learning to generalize semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4340–4350.

[16] E. Tzeng, J. Hoffman, K. Saenko, and T. Darrell, "Adversarial discriminative domain adaptation," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 2962–2971.

[17] M. Mancini, L. Porzi, S. R. Buló, B. Caputo, and E. Ricci, "Inferring latent domains for unsupervised deep domain adaptation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 2, pp. 485–498, Feb. 2021.

[18] Y.-H. Tsai, W.-C. Hung, S. Schulter, K. Sohn, M.-H. Yang, and M. Chandraker, "Learning to adapt structured output space for semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2018, pp. 7472–7481.

[19] Y. Zou, Z. Yu, B. V. K. V. Kumar, and J. Wang, "Unsupervised domain adaptation for semantic segmentation via class-balanced self-training," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2018, pp. 289–305.

[20] J.-Y. Zhu, T. Park, P. Isola, and A. A. Efros, "Unpaired image-to-image translation using cycle-consistent adversarial networks," in *Proc. IEEE Int. Conf. Comput. Vis. (ICCV)*, Oct. 2017, pp. 2242–2251.

[21] Y. Xu, P. Ghamisi, and Y. Avrithis, "Multi-target unsupervised domain adaptation for semantic segmentation without external data," 2024, *arXiv:2405.06502*.

[22] M. Mancini, S. R. Bulò, B. Caputo, and E. Ricci, "AdaGraph: Unifying predictive and continuous domain adaptation through graphs," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 6561–6570.

[23] A. Robey, G. J. Pappas, and H. Hassani, "Model-based domain generalization," in *Proc. Adv. Neural Inf. Process. Syst.*, 2021, pp. 20210–20229.

[24] Q. Zhao, S. Lyu, H. Zhao, B. Liu, L. Chen, and G. Cheng, "Self-training guided disentangled adaptation for cross-domain remote sensing image semantic segmentation," *Int. J. Appl. Earth Observ. Geoinf.*, vol. 127, Mar. 2024, Art. no. 103646.

[25] M. Luo and S. Ji, "Cross-spatiotemporal land-cover classification from VHR remote sensing images with deep learning based domain adaptation," *ISPRS J. Photogramm. Remote Sens.*, vol. 191, pp. 105–128, Sep. 2022.

[26] O. Tasar, A. Giros, Y. Tarabalka, P. Alliez, and S. Clerc, "DAugNet: Unsupervised, multisource, multitarget, and life-long domain adaptation for semantic segmentation of satellite images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 2, pp. 1067–1081, Feb. 2021.

[27] F. Pitie and A. Kokaram, "The linear Monge–Kantorovitch linear colour mapping for example-based colour transfer," in *Proc. IET 4th Eur. Conf. Vis. Media Prod. (CVMP)*, vol. 2007, 2007, p. 23.

[28] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "ADVENT: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2019, pp. 2512–2521.

[29] J. Chen et al., "Memory-contrastive unsupervised domain adaptation for building extraction of high-resolution remote sensing imagery," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5605615.

[30] W. Li, H. Gao, Y. Su, and B. M. Momanyi, "Unsupervised domain adaptation for remote sensing semantic segmentation with transformer," *Remote Sens.*, vol. 14, no. 19, p. 4942, Oct. 2022.

[31] M. Kim and H. Byun, "Learning texture invariant representation for domain adaptation of semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2020, pp. 12972–12981.

[32] Y. Zhao, P. Guo, Z. Sun, X. Chen, and H. Gao, "ResiDualGAN: Resize-residual DualGAN for cross-domain remote sensing images semantic segmentation," *Remote Sens.*, vol. 15, no. 5, p. 1428, 2023.

[33] Y. Zhang et al., "Domain enhanced arbitrary image style transfer via contrastive learning," in *Proc. Special Interest Group Comput. Graph. Interact. Techn. Conf.*, Aug. 2022, pp. 1–8.

[34] B. Jähne, *Digital Image Processing*. Cham, Switzerland: Springer, 2005.

[35] J. Yoo, Y. Uh, S. Chun, B. Kang, and J.-W. Ha, "Photorealistic style transfer via wavelet transforms," in *Proc. IEEE/CVF Int. Conf. Comput. Vis. (ICCV)*, Oct. 2019, pp. 9035–9044.

[36] J. Liu, W. Xuan, Y. Gan, Y. Zhan, J. Liu, and B. Du, "An end-to-end supervised domain adaptation framework for cross-domain change detection," *Pattern Recognit.*, vol. 132, Dec. 2022, Art. no. 108960.

[37] Y. Tian, X. Zhao, and W. Huang, "Meta-learning approaches for learning-to-learn in deep learning: A survey," *Neurocomputing*, vol. 494, pp. 203–223, Jul. 2022.

[38] T. Hospedales, A. Antoniou, P. Micaelli, and A. Storkey, "Meta-learning in neural networks: A survey," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 44, no. 9, pp. 5149–5169, Sep. 2021.

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[39] A. Santoro, S. Bartunov, M. Botvinick, D. Wierstra, and T. Lillicrap, "Meta-learning with memory-augmented neural networks," in *Proc. Int. Conf. Mach. Learn.*, Jun. 2016, pp. 1842–1850.
[40] A. Graves, G. Wayne, and I. Danihelka, "Neural Turing machines," 2014, *arXiv:1410.5401*.
[41] S. Sukhbaatar, A. Szlam, J. Weston, and R. Fergus, "End-to-end memory networks," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 28, Dec. 2015, pp. 2440–2448.
[42] J. Su, H. Shen, L. Peng, and D. Hu, "Few-shot domain-adaptive anomaly detection for cross-site brain images," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 3, pp. 1819–1835, Mar. 2021.
[43] J. Su et al., "M_2DC: A meta-learning framework for generalizable diagnostic classification of major depressive disorder," *IEEE Trans. Med. Imag.*, vol. 44, no. 2, pp. 855–867, Feb. 2025.
[44] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2016, pp. 770–778.
[45] L.-C. Chen, Y. Zhu, G. Papandreou, F. Schroff, and H. Adam, "Encoder-decoder with Atrous separable convolution for semantic image segmentation," in *Proc. Eur. Conf. Comput. Vis. (ECCV)*, Jan. 2018, pp. 801–818.
[46] P. Isola, J.-Y. Zhu, T. Zhou, and A. A. Efros, "Image-to-image translation with conditional adversarial networks," in *Proc. IEEE Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jul. 2017, pp. 5967–5976.
[47] Y. Wang et al., "Semi-supervised semantic segmentation using unreliable pseudo-labels," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2022, pp. 4238–4247.
[48] J. Xia, N. Yokoya, B. Adriano, and C. Broni-Bediako, "OpenEarthMap: A benchmark dataset for global high-resolution land cover mapping," in *Proc. IEEE/CVF Winter Conf. Appl. Comput. Vis. (WACV)*, Jan. 2023, pp. 6243–6253.
[49] J. Wang, Z. Zheng, A. Ma, X. Lu, and Y. Zhong, "LoveDA: A remote sensing land-cover dataset for domain adaptive semantic segmentation," 2021, *arXiv:2110.08733*.
[50] E. Maggiori, Y. Tarabalka, G. Charpiat, and P. Alliez, "Can semantic labeling methods generalize to any city? The inria aerial image labeling benchmark," in *Proc. IEEE Int. Geosci. Remote Sens. Symp. (IGARSS)*, Jul. 2017, pp. 3226–3229.
[51] *Open Cities AI Competition: Segmenting Buildings for Disaster Resilience*. Accessed: Mar. 2020. [Online]. Available: https://www.drivendata.org/competitions/60/buildingsegmentation-disaster-resilience/
[52] G. Baier, A. Deschemps, M. Schmitt, and N. Yokoya, "Synthesizing optical and SAR imagery from land cover maps and auxiliary raster data," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 4701312.
[53] A. Boguszewski, D. Batorski, N. Ziemba-Jankowska, T. Dziedzic, and A. Zambrzycka, "LandCover.Ai: Dataset for automatic mapping of buildings, woodlands, water and roads from aerial imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2021, pp. 1102–1110.
[54] R. Gupta et al., "Creating xBD: A dataset for assessing building damage from satellite imagery," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops*, Jun. 2019, pp. 10–17.
[55] M. Bińkowski, D. J. Sutherland, M. Arbel, and A. Gretton, "Demystifying MMD GANs," 2018, *arXiv:1801.01401*.
[56] M. Heusel, H. Ramsauer, T. Unterthiner, B. Nessler, and S. Hochreiter, "GANs trained by a two time-scale update rule converge to a local Nash equilibrium," in *Proc. Adv. Neural Inf. Process. Syst.*, Jan. 2017, pp. 6629–6640.
[57] T. Salimans, I. Goodfellow, W. Zaremba, V. Cheung, A. Radford, and X. Chen, "Improved techniques for training GANs," in *Proc. Adv. Neural Inf. Process. Syst.*, vol. 29, 2016, pp. 1–7.
[58] A. Mittal, R. Soundararajan, and A. C. Bovik, "Making a 'completely blind' image quality analyzer," *IEEE Signal Process. Lett.*, vol. 20, no. 3, pp. 209–212, Apr. 2012.
[59] T. Zhang, F. Gao, J. Dong, and Q. Du, "Remote sensing image translation via style-based recalibration module and improved style discriminator," *IEEE Geosci. Remote Sens. Lett.*, vol. 19, pp. 1–5, 2022.



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